

Answer: B

$$23 \text{ fringe separation} = \frac{\lambda D}{d} = \frac{(600 \times 10^{-9})(2.5)}{0.40 \times 10^{-3}} = 3.75 \times 10^{-3} \text{ m}$$

$$x = 5 \times 3.75 \times 10^{-3} \text{ m} = 1.9 \times 10^{-2} \text{ m}$$

Answer: D

$$24 \quad d \sin \theta = n \lambda$$